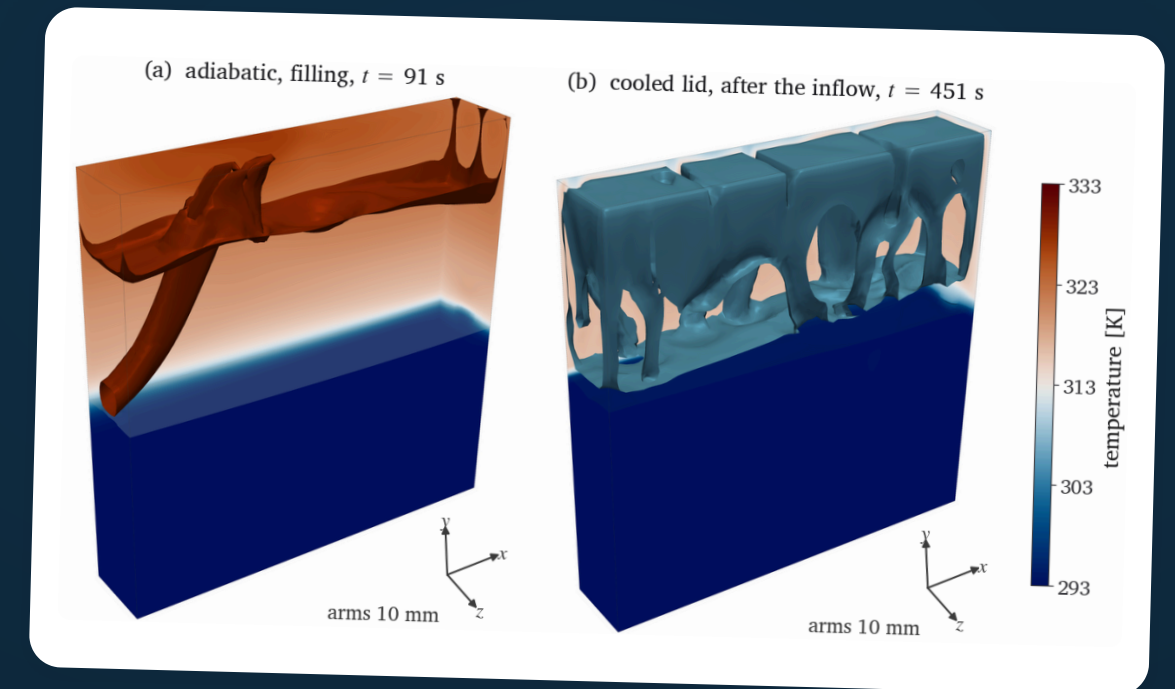


Data-based Numerical Simulation of Buoyant Jet Flow

What a recurrence replay loses when the scalar drives the flow, and how much of it can be repaired

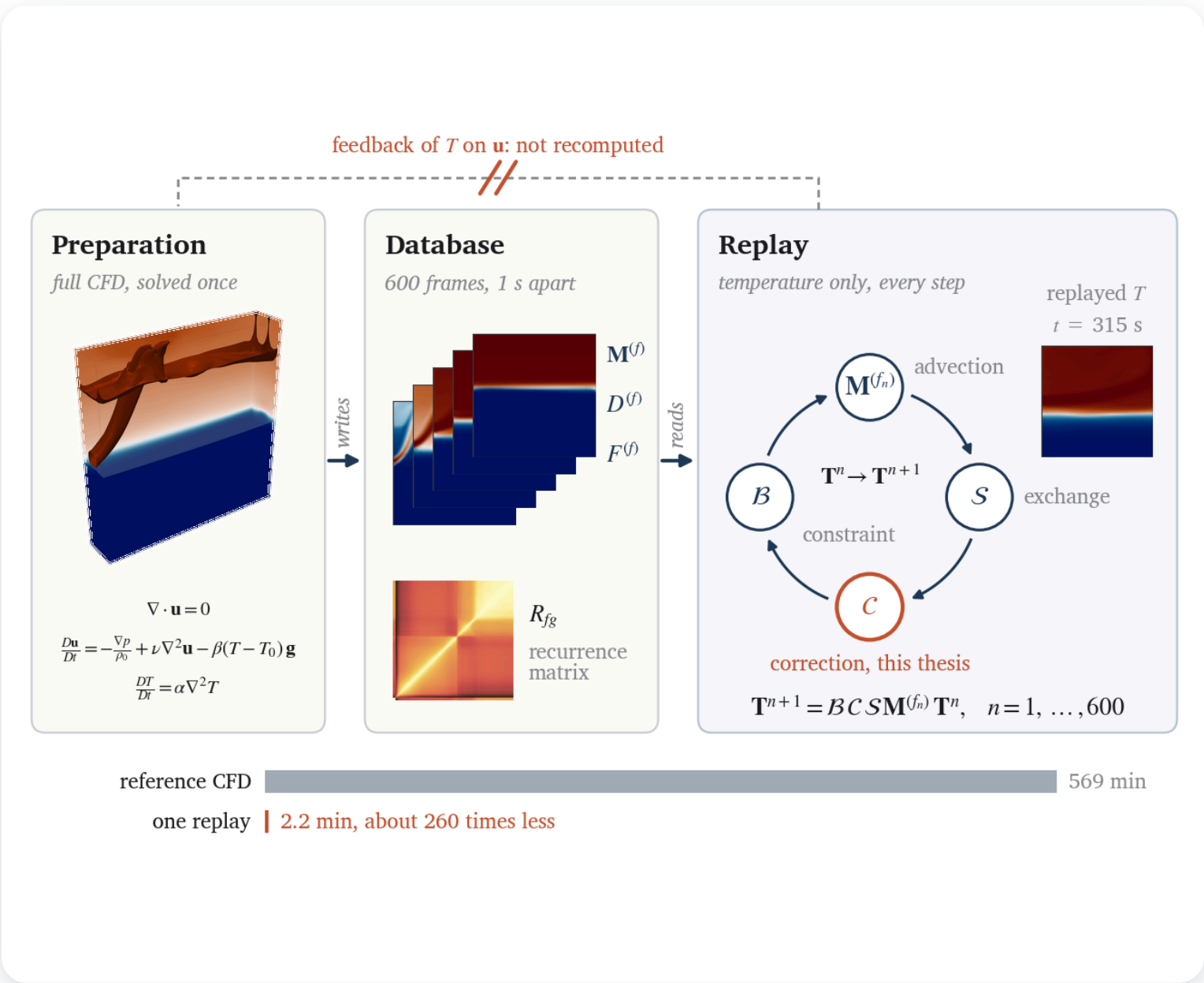
MSc thesis defense

Supervisor: Prof. Stefan Pirker · Department of Particulate Flow Modelling



THE METHOD

Simulate the flow once. Replay it many times.



A full CFD run records **how the fluid moves**, frame by frame, as cell-to-cell maps.

The replay applies these maps to the temperature again and again. Momentum and pressure are **never solved again**.

260×

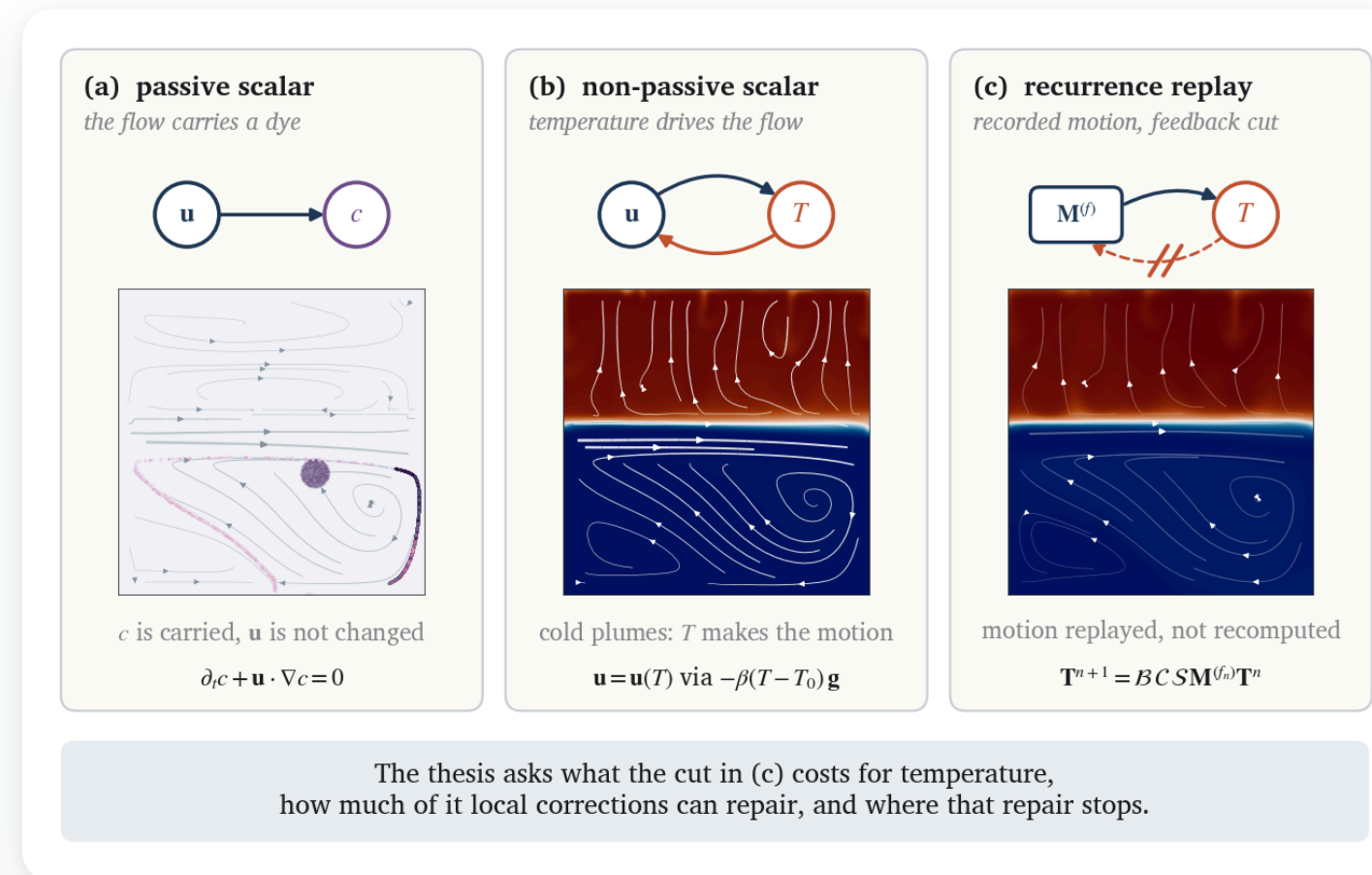
cheaper on average

569 → 1–5

minutes per run

THE QUESTION

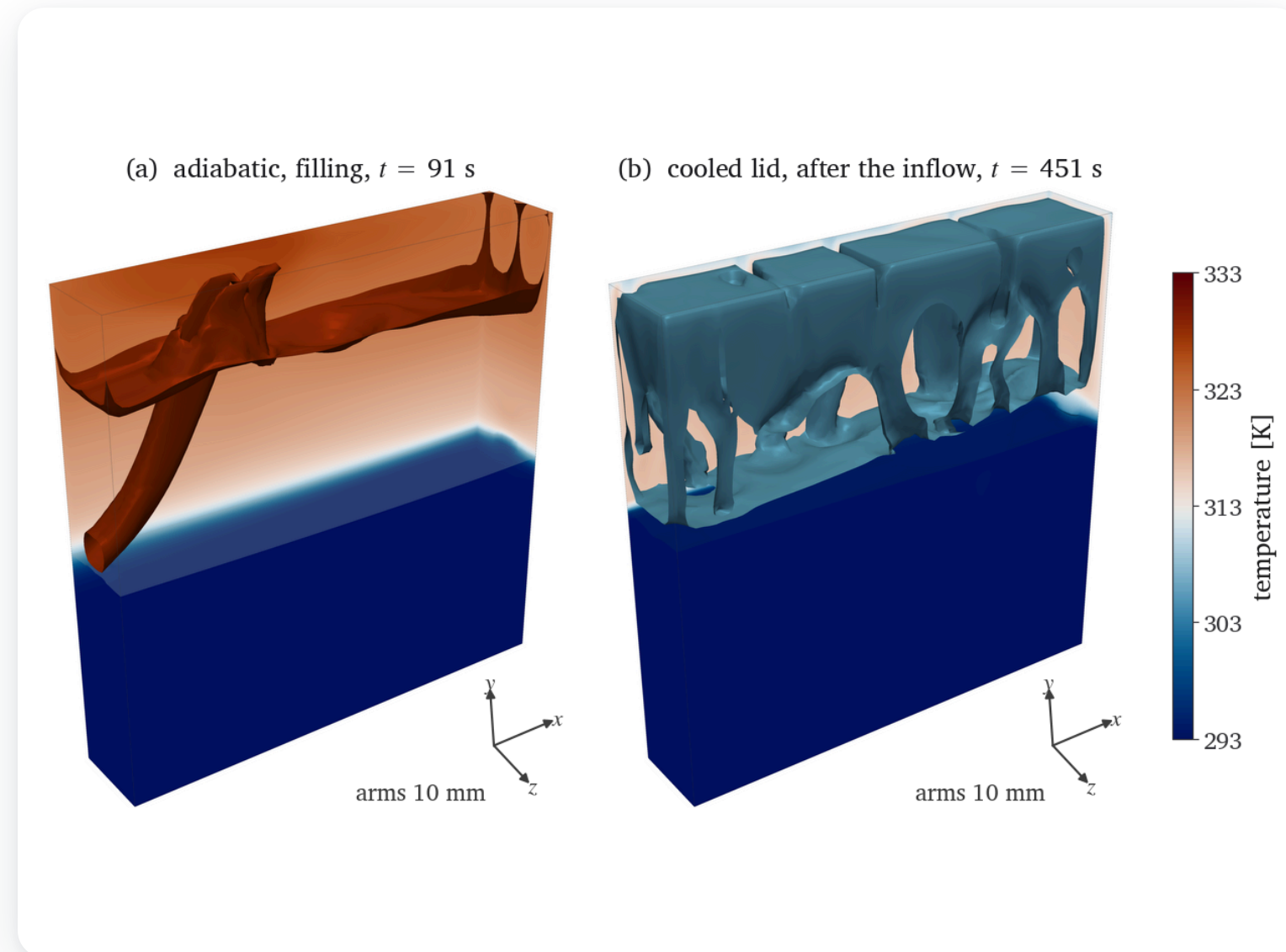
**rCFD was built for a scalar that only rides the flow.
Temperature pushes back.**



What does the replay lose when the scalar drives the flow, and how much can be repaired
without solving the flow again?

THE TEST CASE

A small tank, a warm jet, and a front that must stay sharp



75×75×15 mm tank

1 mm cubic cells

warm inflow **333 K** for **315 s**

Re $\approx$ 99 laminar

Ri 1.84 buoyancy wins

600 recorded frames

Two lids: **adiabatic** and **cooled** ($h = 100 \text{ W m}^{-2} \text{ K}^{-1}$).

Reference: ANSYS Fluent, Boussinesq flow. Replay: the rCFD library of the department.

Three claims and one open point

1

One knob, two answers

No single mixing constant is best, and the reason is physical.

2

Sharpen safely

A bounded anti-diffusive correction rebuilds the front with no overshoot.

3

The missing cooling

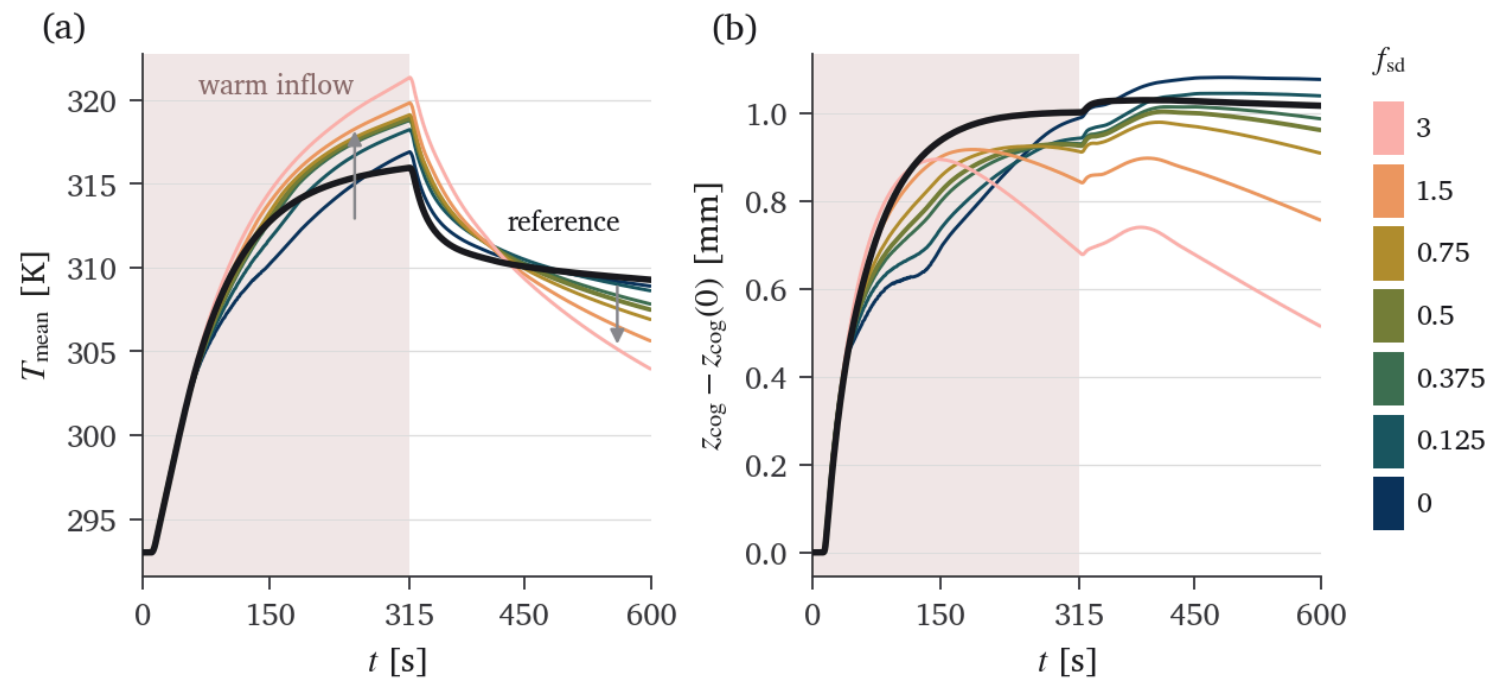
Most of it was a material mismatch, not a transport failure.

4

What remains

A transport defect that only recorded transport can fix.

One mixing knob, two different best values



The replay adds mixing with a face exchange of strength f_{sd} . Over seven values:

- **Mean temperature** is best with little mixing (RMSE 1.07 K).
- **Height of the heat** is best near $f_{sd} = 0.5$.
- At the largest strength the tank drains **5.08 K** too much.

The outlet sits just below the front. A smeared front leaks heat.

A wider front puts warm fluid at the height of the outlet.

+2.7 K

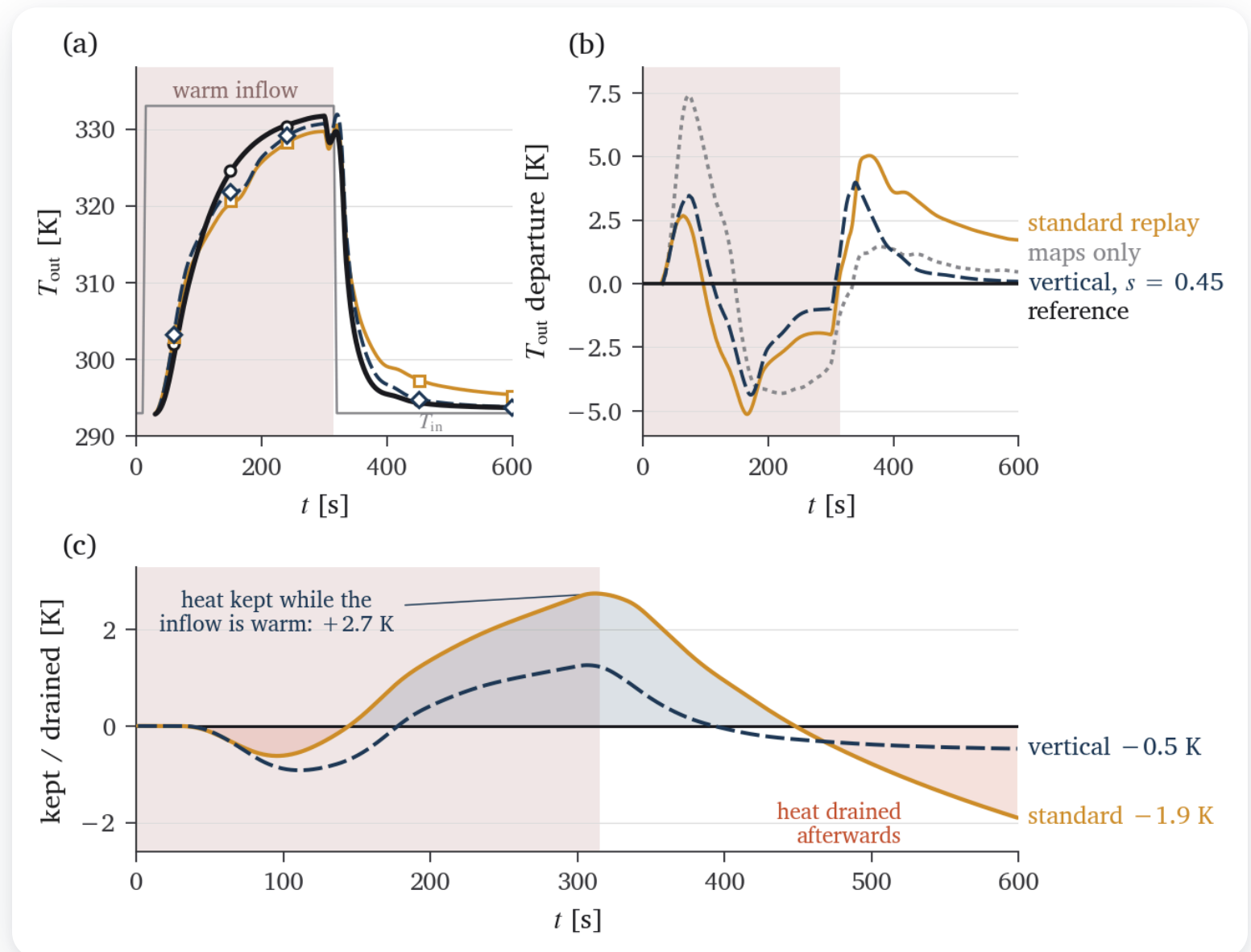
kept in the tank
at 315 s

−1.9 K

drained by
the end

With the correction: **−0.5 K**.

A numerical error at a front becomes an **energy error** wherever an outlet sits near it.

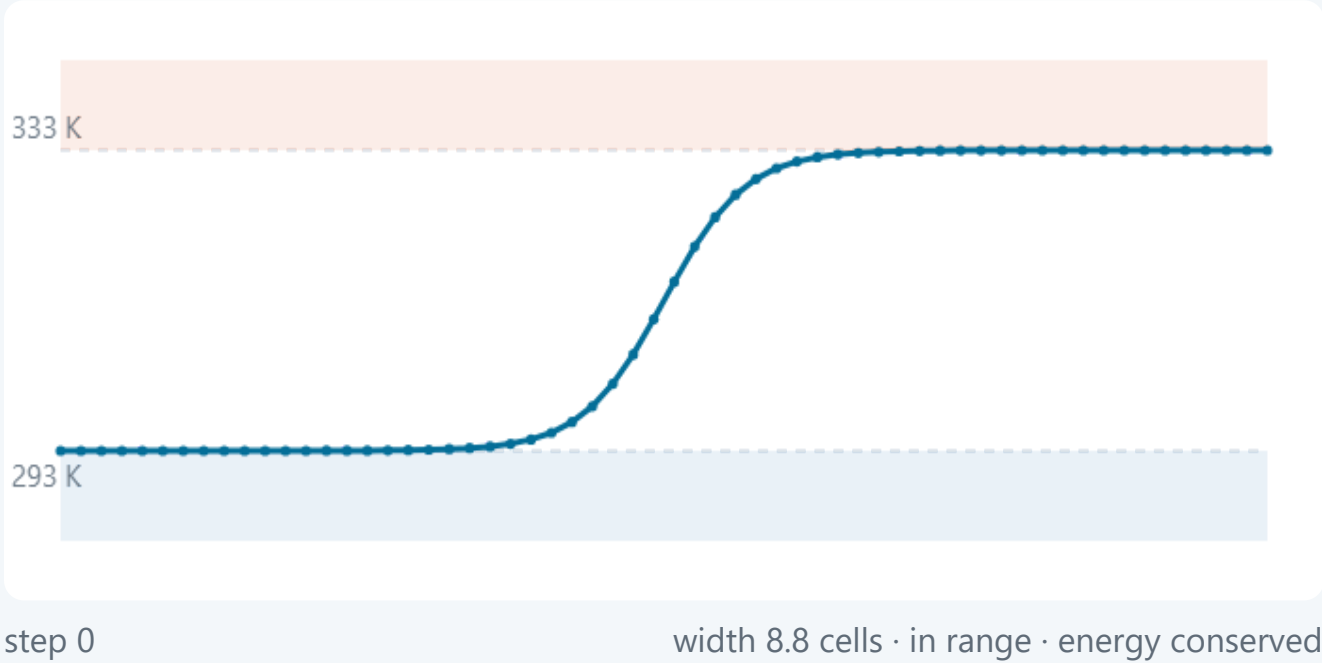


Anti-diffusion explodes. A local-bounds limiter makes it safe.

Negative diffusion, $\nu = -0.1$



Limited correction, $s = 0.25$

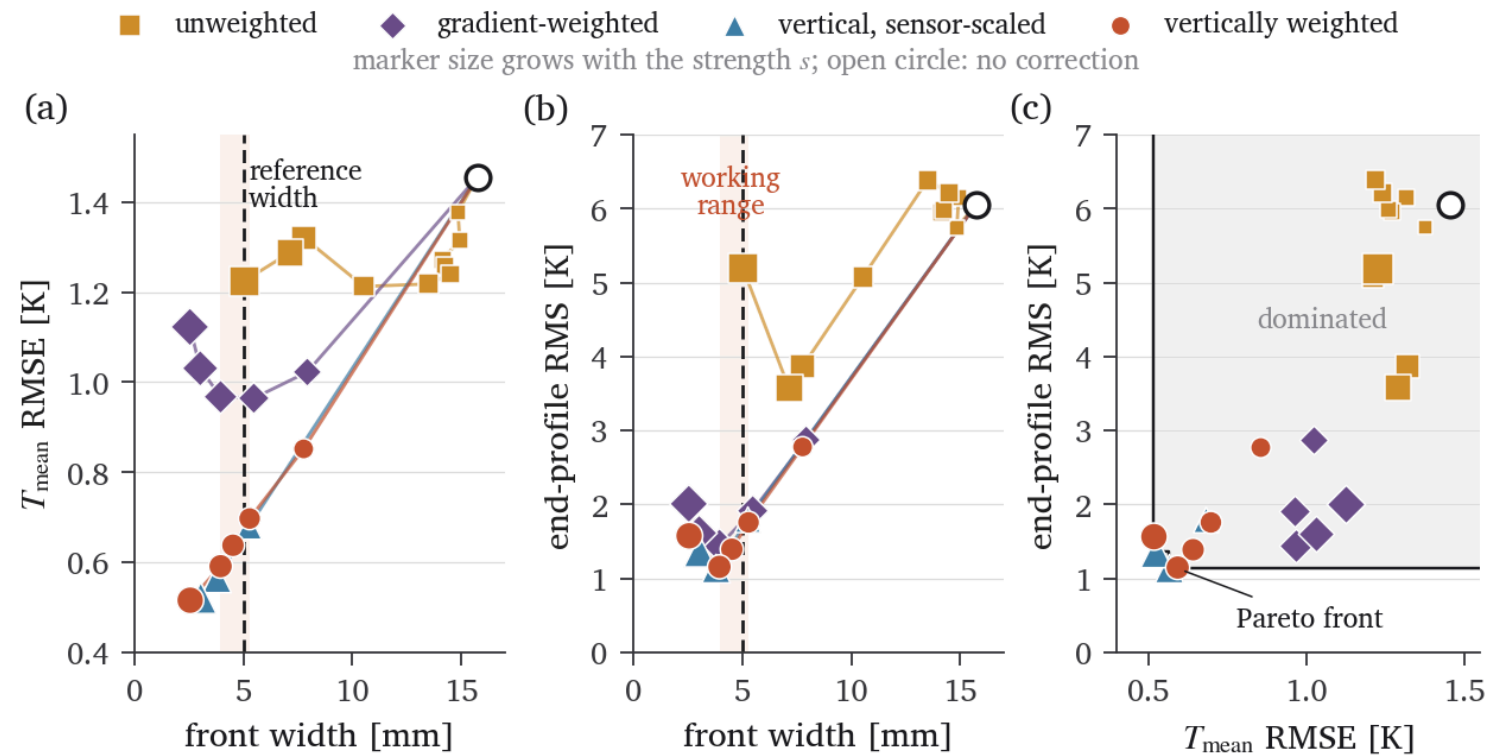


▶ Run

Reset

$\Delta q_f = s \cdot \min(Q_h^+, Q_c^-) \cdot \text{conservative} \cdot \text{no new extremum} \cdot \text{no clamp needed for } s \leq 1/6$

Act across the front, not along it



Each face gets a weight $\cos^2$ of its angle to the vertical: the face coefficient of a rank-one diffusion tensor along the front normal.

- Only the **vertical** form lowers both errors as the front reaches 5.0 mm.
- Unweighted, it needs $s = 0.9$ and leaves a saw-toothed front 3.1 mm too high.

The front comes back, and nothing leaves the range of the data

15.7 → 5.3 / 4.5 mm

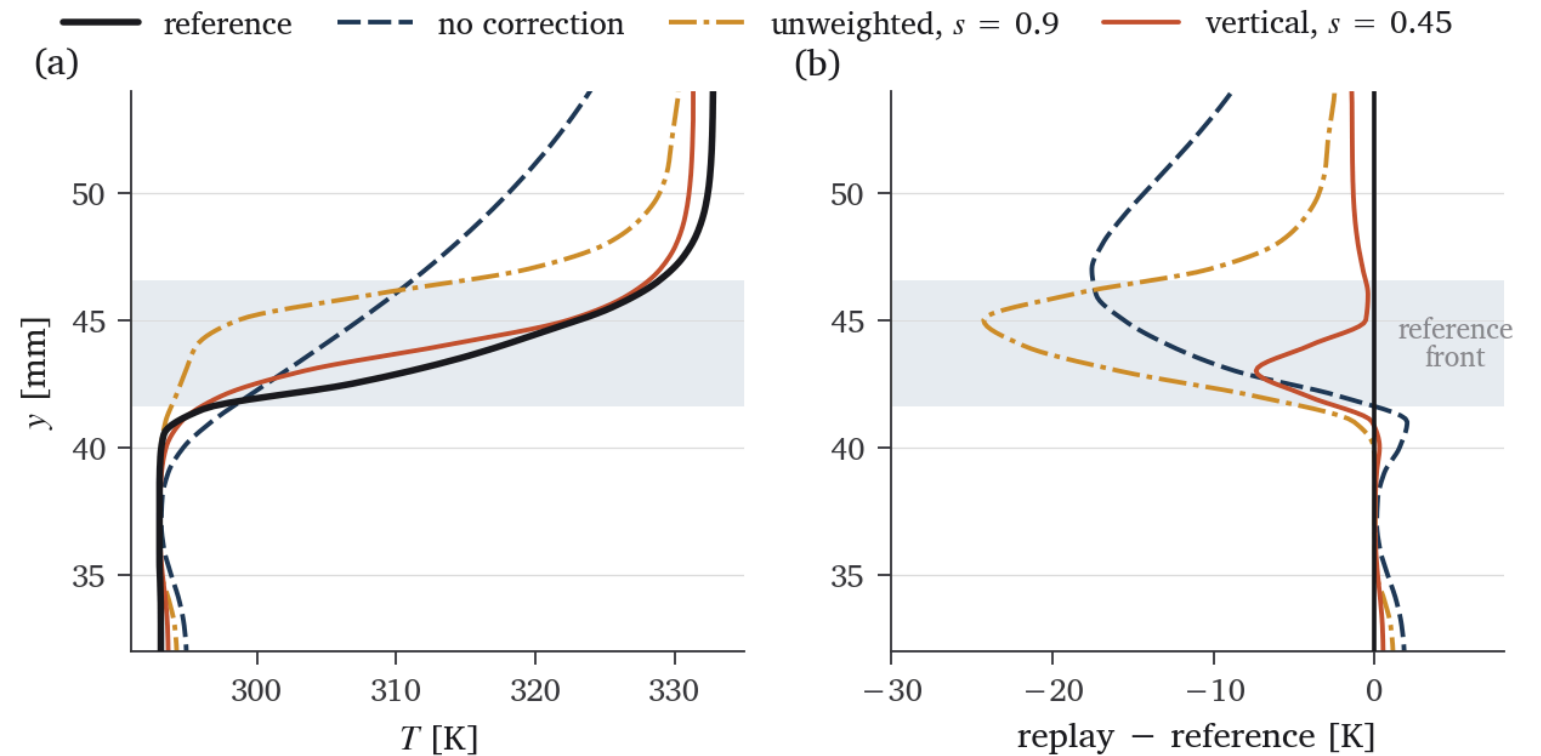
front width at $s = 0.40 / 0.45$ ·
reference 5.0 mm

1.46 → 0.64 K 6.05 → 1.39 K

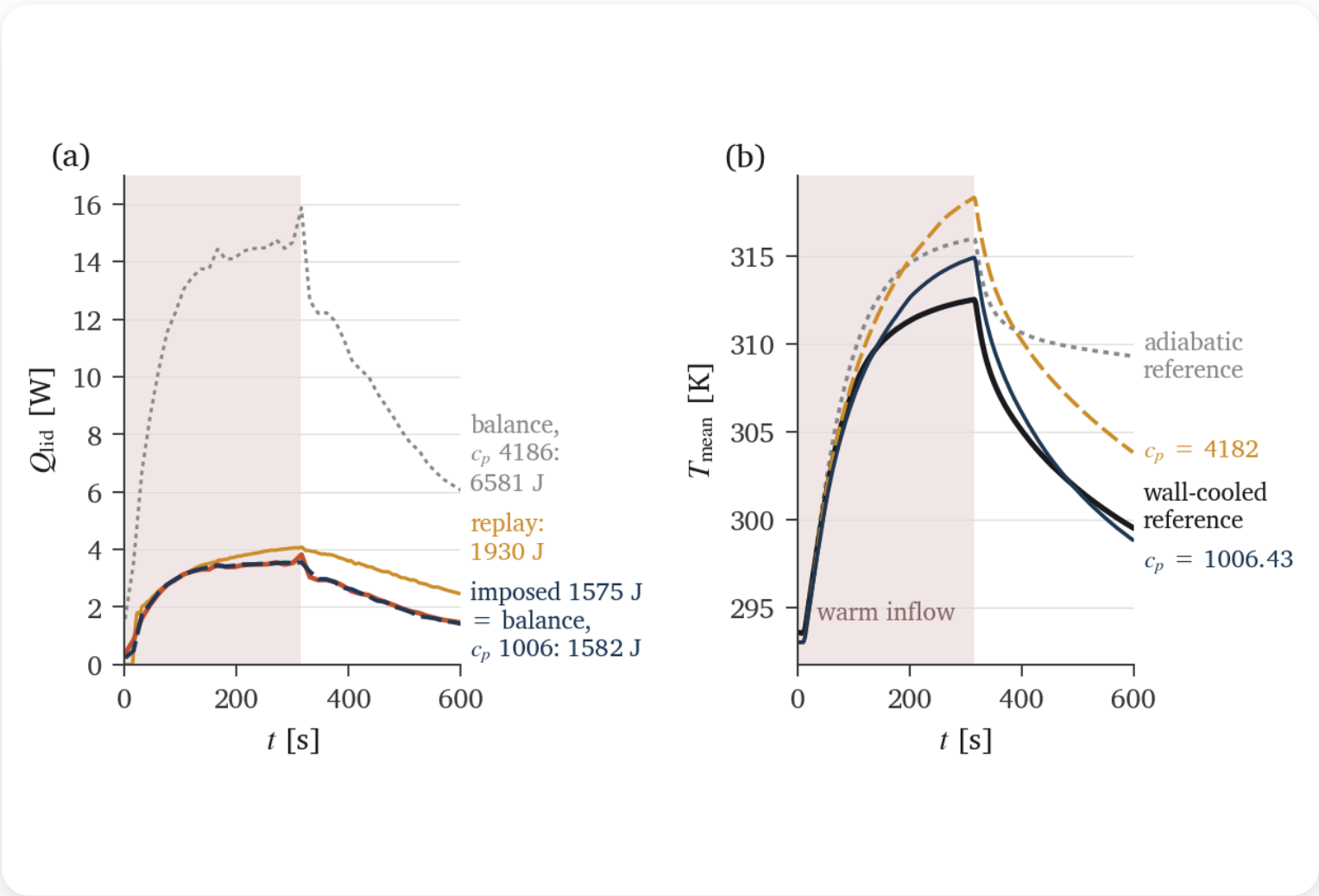
trace error

end profile

No end field in any run left the range of the data.



The replay stayed 4.34 K too warm. The budget showed why.



I traced every heat term to the specific heat it was computed with.

REFERENCE

"water" with c_p of **air**
1006.43 J kg⁻¹K⁻¹

REPLAY

water
4182 J kg⁻¹K⁻¹

× **4.155**
same wall power, four times less
cooling in the replay

On one fluid, the replay follows the reference

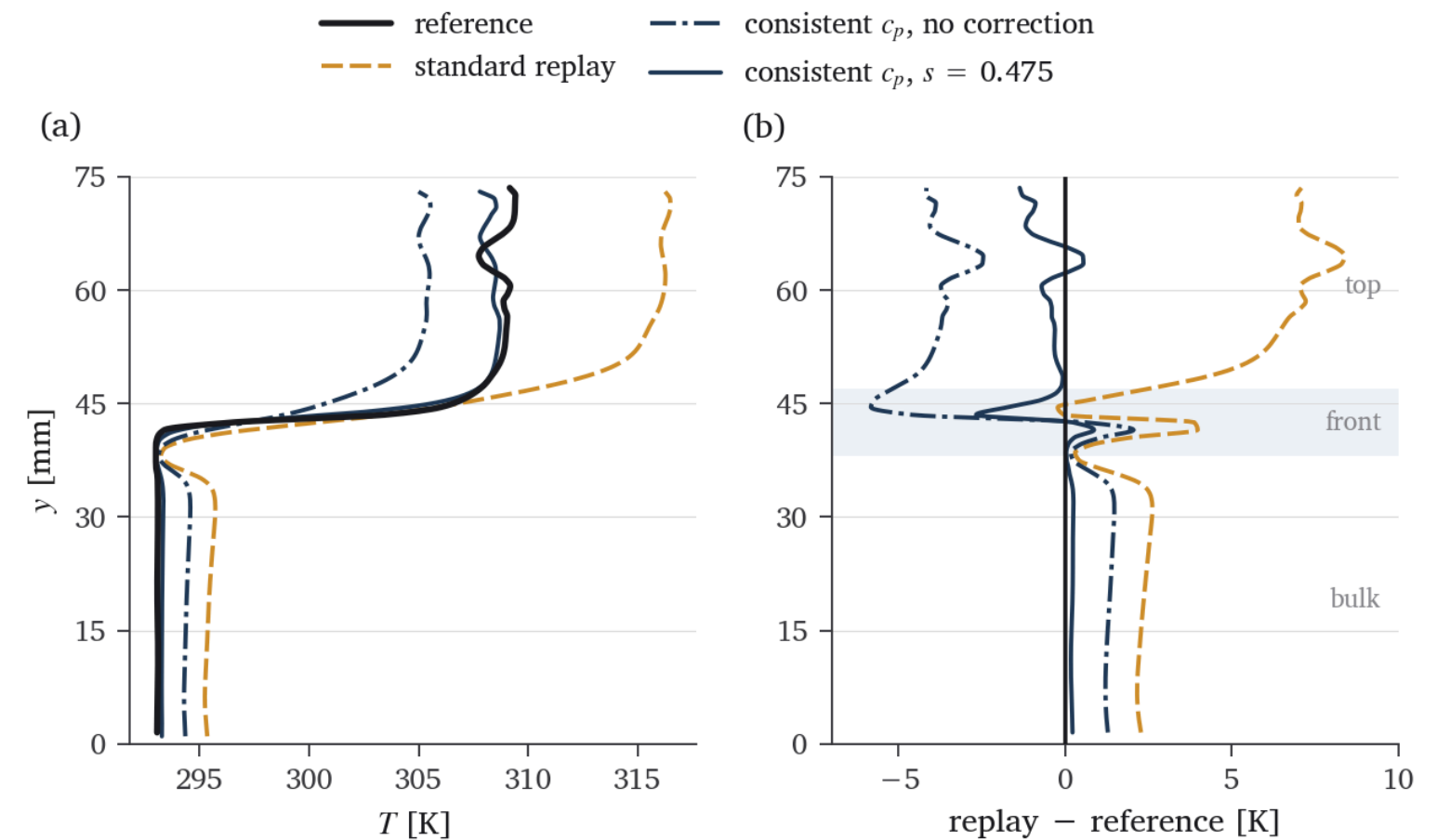
4.34 → 0.66 K

end error, consistent fluid,
declared coefficient

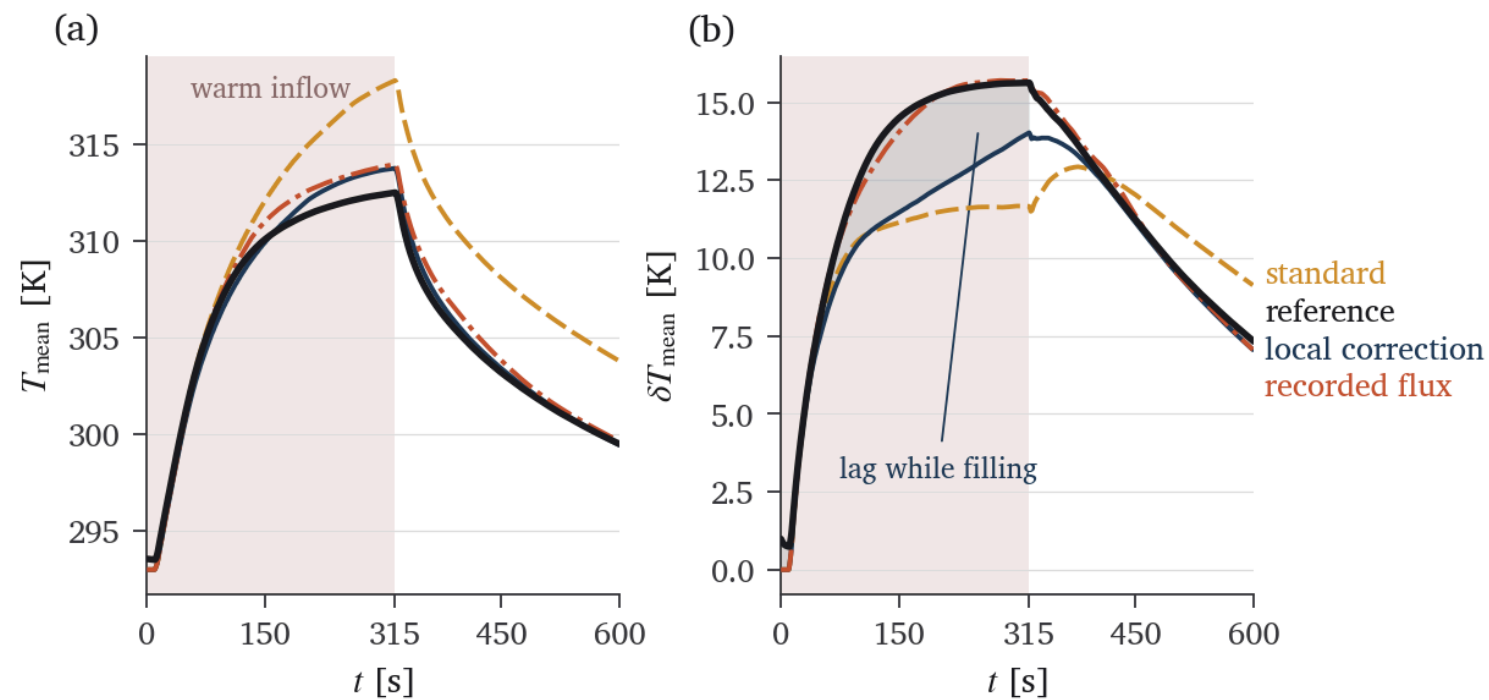
0.06 K

with the correction at $s \approx 0.55$, on
9.62 K of cooling

End profile within **0.41 K**; all three zones within 0.2 K.



One defect survives every local repair: it is transport



While the tank fills, the replay separates its layers too slowly. A conservative local rule cannot create missing transport.

A vertical convective flux read from the recording, $F = \rho c_p v (T - \bar{T})$, supplies it:

$$1.66 \rightarrow 0.26 \text{ K}$$

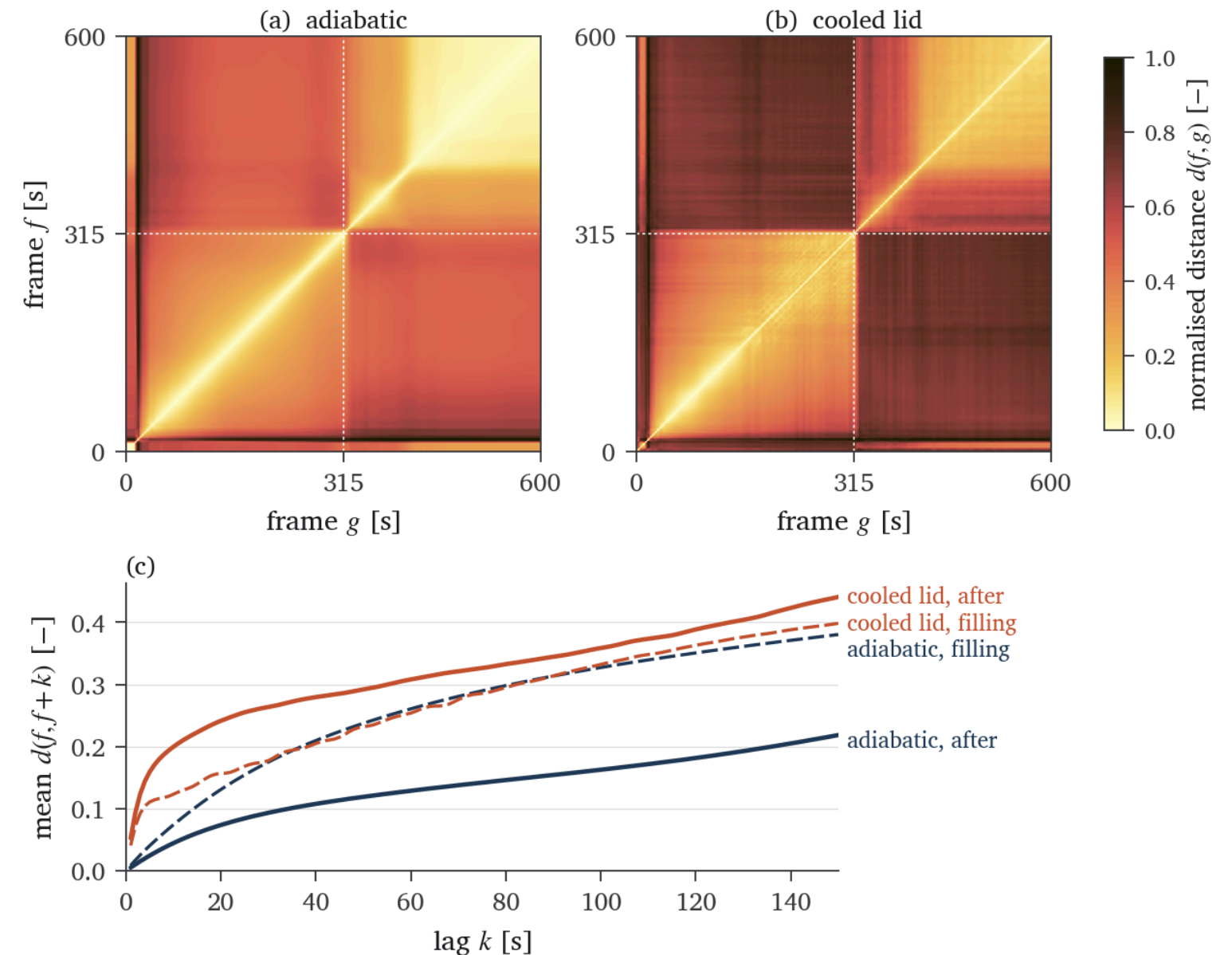
heterogeneity error

Limitation: the flux is tied to the run it was recorded from.

A recording belongs to a flow regime, not to a case

After the inflow, the adiabatic flow hardly changes.
Under the cooled lid, cold plumes keep renewing it.

A boundary condition switched the regime by itself. In ML words: the new case is **out of distribution**, and no coefficient recovers a motion the recording never saw.



Three things this tank teaches beyond itself

1

Regime, not case

A recurrence database belongs to a flow regime; a boundary condition can switch the regime by itself.

2

Front → budget

A numerical error at a front becomes an energy error near an outlet; a bounded anti-diffusive correction repairs both.

3

Same fluid, same clock

Two simulations can only be compared on the same fluid and at the same instant.

Next: an implicit exchange so the recorded diffusivity can act · a recorded wall coefficient per cell · multi-state databases.

Thank you.

Questions are welcome.

Four operators per step

$$T^{n+1} = B \left(C \left(S \left(M^{(f_n)} T^n \right) \right) \right)$$

M · map

$T_j^* = \sum_i M_{ji} T_i$, columns sum to 1

S · face swap

$\Delta = f_{sd} V (T_1 - T_0)/2$; $D_{eq} = f_{sd} c_s^2/(2\Delta t)$

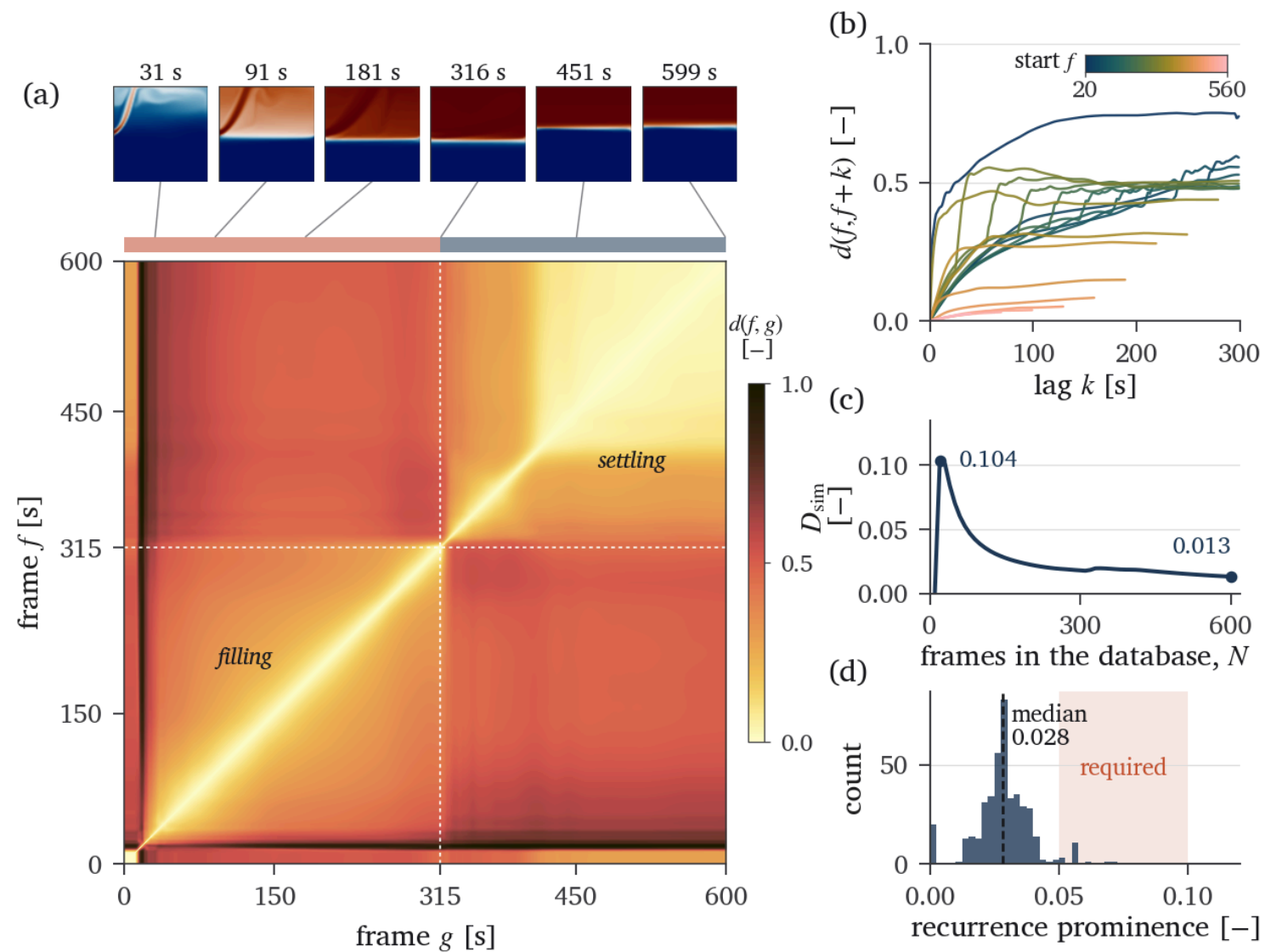
C · correction

$\Delta q = s \cdot \min(Q_h^+, Q_c^-)$,
weighted $\cos^2$

B · balance

global conservation constraint
(mass-acting swaps)

The buoyant jet is not recurrent: the replay runs in sequence



Two blocks: filling and settling. No state comes back.

Median prominence **0.028**; only 4 % of the vectors reach 0.05 (required 0.05–0.1).

So frame f of the replay uses the map of frame f : a sequential re-transport, as in Dabbagh et al. 2020.

Why the recorded diffusivity acted like a constant

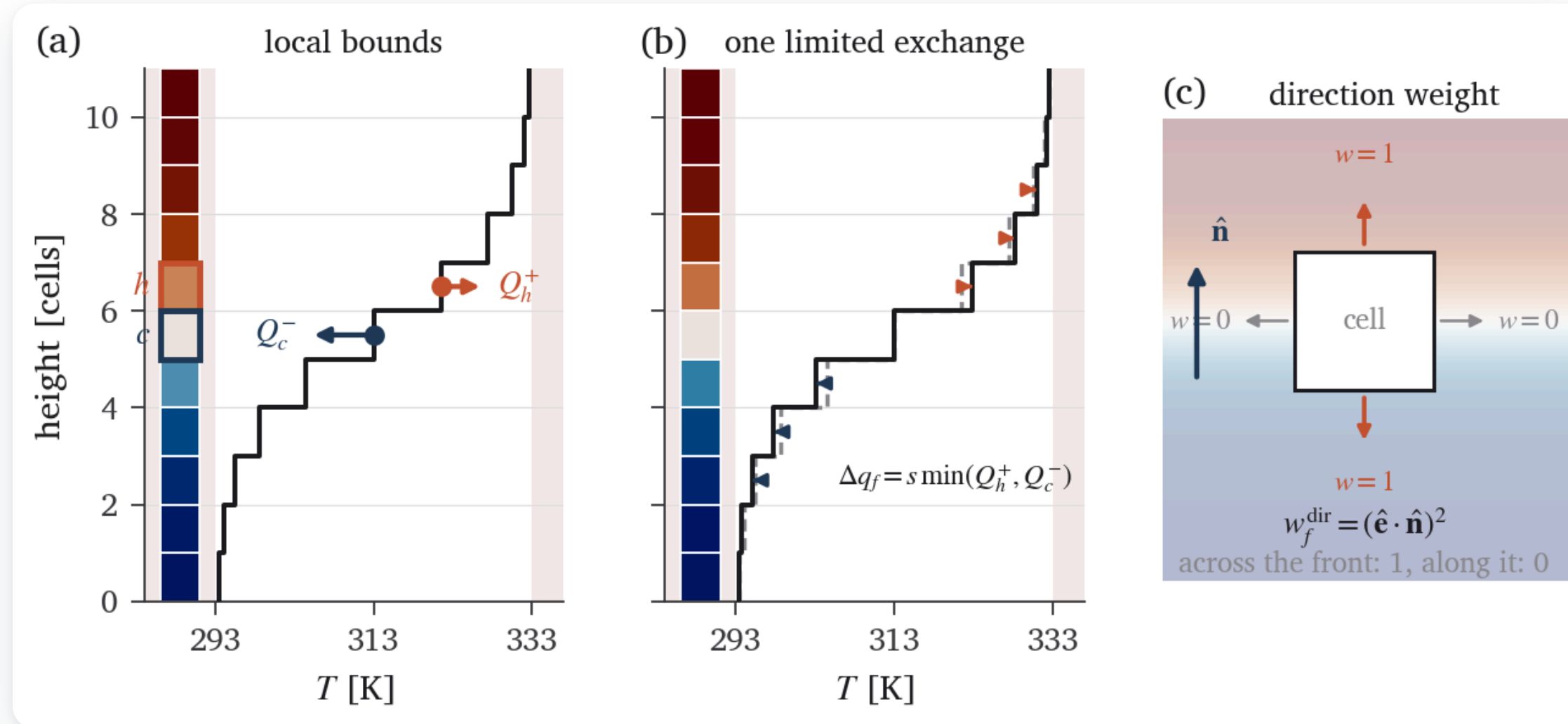
The exchange is explicit and capped at half a cell per face and step:

$$Di = \frac{D\Delta t}{c_s^2} \leq 0.5$$

The recorded field asks for $Di \approx 4$ in every cell (smallest value $7.2 \cdot 10^{-7} \text{ m}^2/\text{s} > 5 \cdot 10^{-7} \text{ m}^2/\text{s}$). Every face hits the cap, so it behaves like $f_{sd} = 0.5$: three replays are bit-identical.

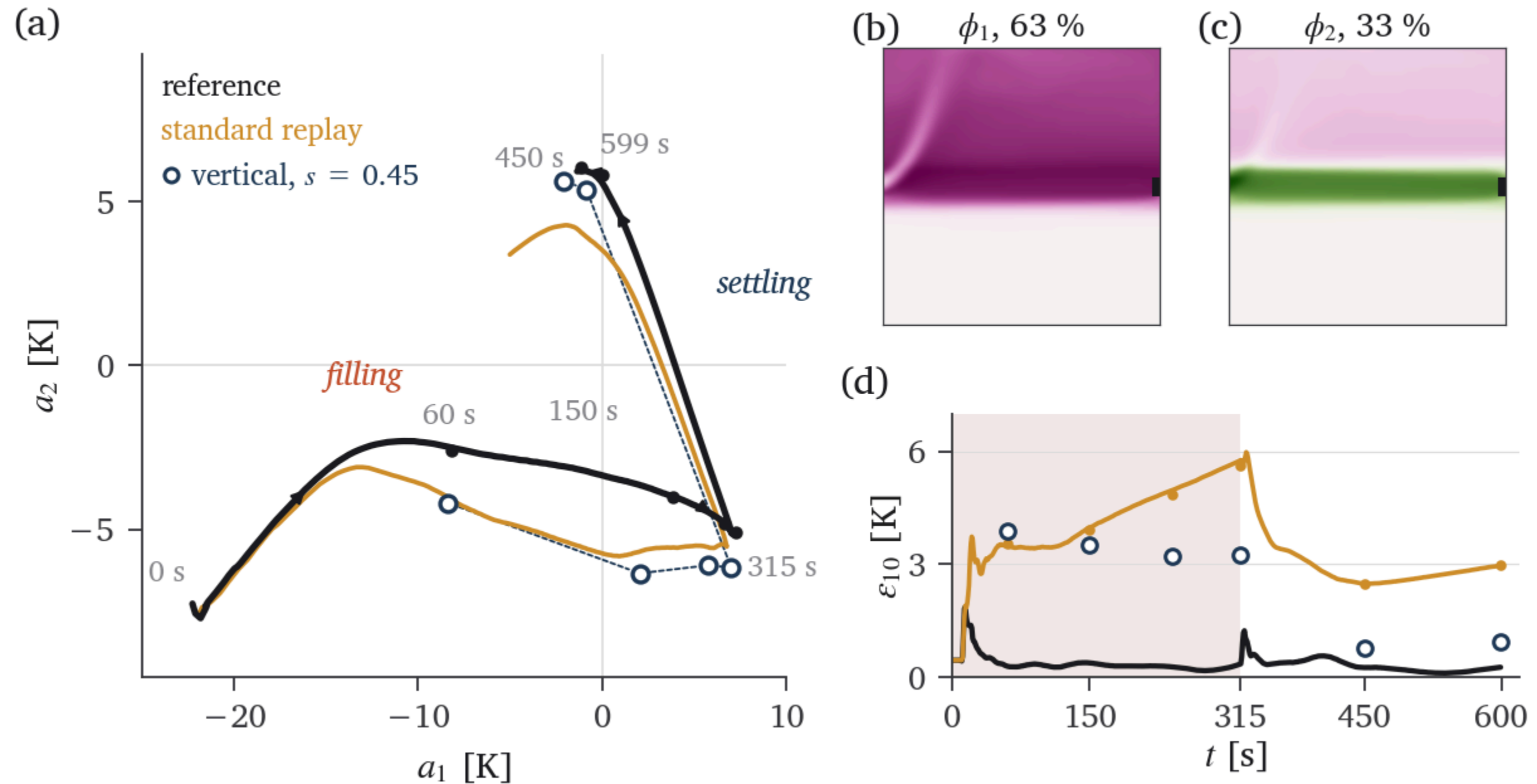
Fix: a shorter replay step or an implicit exchange.

The correction operator on a model front

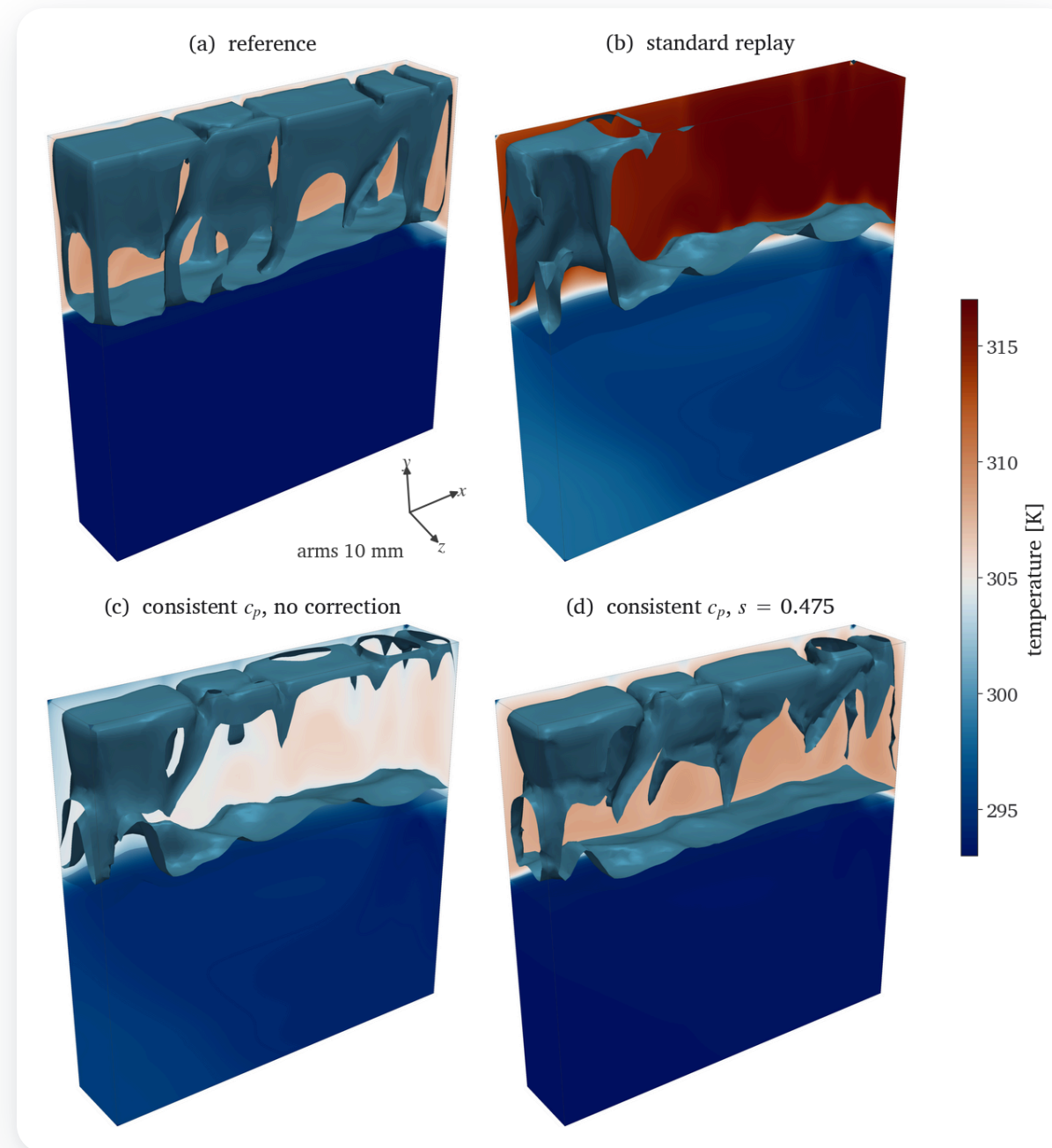


$Q_h^+ = C_h(T_h^{\text{max}} - T_h)$, $Q_c^- = C_c(T_c - T_c^{\text{min}})$, $\Delta q = s \cdot \min(Q_h^+, Q_c^-)$; for $s \leq 1/n_f = 1/6$ no clamp acts.

The corrected replay returns to the recorded subspace



Only the corrected replay carries plumes across the layer



REFERENCES

Main references

- [1] Lichtenegger, T., Pirker, S. (2016). Recurrence CFD, a novel approach to simulate multiphase flows with strongly separated time scales. *Chem. Eng. Sci.*
- [2] Pirker, S., Lichtenegger, T. (2018). Efficient time-extrapolation of single- and multiphase simulations by transport based recurrence CFD. *Chem. Eng. Sci.*
- [3] Lichtenegger, T. et al. (2019). Multi-state recurrence CFD.
- [4] Dabbagh, F. et al. (2020). Sequential recurrence path in a fluidised bed.
- [5] Boris, J. P., Book, D. L. (1973). Flux-corrected transport. *J. Comput. Phys.*
- [6] Zalesak, S. T. (1979). Fully multidimensional flux-corrected transport. *J. Comput. Phys.*
- [7] Patankar, S. V. (1980). *Numerical Heat Transfer and Fluid Flow*.